

## **DIGITAL FORENSIC EXAMINATION REPORT**

### *Chrome Web-Browser History Artifact Analysis*

**Case Reference: CIP-B105-CS1-C11-26-DFIT-173065**

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**CLASSIFICATION: Training / Laboratory Exercise Not a Live Legal Proceeding**  
*For educational use only. Evidence items are course-supplied Chrome History artifacts.*

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## 1. Executive Summary

I examined a Google Chrome History SQLite artifact named History\_main, together with six related reference and comparison Chrome History databases supplied for this laboratory exercise. I carried out the examination on a Kali Linux forensic workstation using sqlite3, Python 3, and standard hashing tools. Copying, hashing, querying, and other examination activities were recorded so that the process could be followed and reproduced.

The primary artifact shows one continuous browsing session on 19 April 2022. The timestamps below are converted from Chrome's stored timestamps and are treated as UTC for this report, subject to the limitation discussed in Section 7. During the session, the browser history shows that the user:

- logged into Craigslist and created a classified advertisement in the Baltimore “health and beauty” category, with posting ID 7473121658 and the title “cheaper than Rx supplements”;
- logged into Gmail using unsub.fscs@gmail.com and exchanged messages with a prospective buyer in a thread connected to the advertisement;
- opened a later email whose subject contained a Bitcoin transaction ID, followed a Gmail link to an Imgur image, and downloaded the image as proof\_of\_payment.png to the Desktop;
- searched for and used Blockchain.com Explorer to check the same transaction ID and a related Bitcoin address; and
- logged into Kraken, viewed a funding/ledger balance, and followed a link from the Kraken interface to mempool.space for the same transaction ID.

One important integrity issue was identified. Chrome\_History\_Cryptocurrency\_Lab.db has exactly the same MD5 and SHA-256 hashes as History\_main, meaning the two files are byte-for-byte identical even though they have different filenames.

Overall, the browsing sequence is consistent with an online marketplace transaction involving cryptocurrency. It could also fit a common payment-verification scam pattern, such as a buyer sending an image presented as proof of payment. Based on the artifacts available, I cannot determine the person's intent or confirm whether the payment was genuine. I therefore present the evidence and the most reasonable interpretation without treating either explanation as proven.

## 2. Authority, Scope and Investigative Questions

### 2.1 Authority

This examination was completed as a supervised DFIT laboratory exercise under course module CIP-B105-CS1-C11-26 (Digital Forensics and Incident Response). The evidence was supplied through the course shared folder (/media/sf\_LabFiles/LabFiles) rather than collected under independent legal authority. I have therefore used a professional forensic-report structure and methodology for training purposes, but this report is not a live investigative or judicial report.

### 2.2 Scope

The examination was limited to seven Chrome History SQLite artifacts: one primary file, History\_main, and six comparison/reference files. These included a purported cryptocurrency lab database, three recreation databases, and a to-database demo file. No browser cache, cookies, downloaded-file contents, email server records, or cryptocurrency-exchange account records were available for examination.

### 2.3 Investigative Questions

- What browsing activity appears in the primary profile, and in what order did it occur?
- Was a Craigslist advertisement created, and what can be established about its title and category?
- What Gmail activity relates to the advertisement, and what did the correspondence lead to?
- What cryptocurrency-related activity appears in the history, and how well can the transaction be corroborated?
- What file was downloaded, and what source and referring page are recorded?
- Do the supplied recreation and lab databases help interpret the primary artifact, and do they create any integrity or chain-of-custody concerns?

## 3. Evidence Inventory and Integrity Verification

I copied all evidence files from the read-only course share into a local evidence/ directory. MD5 and SHA-256 hashes were calculated immediately after copying and are recorded in Appendix C. The primary History\_main file was then copied to working/History\_main.sqlite and hashed again. Its SHA-256 hash matched the evidence copy exactly, confirming that the working copy used for analysis was unchanged.

The most significant integrity issue is that evidence/Chrome\_History\_Cryptocurrency\_Lab.db and evidence/History\_main have identical MD5 and SHA-256 values. This means they are not simply similar

databases; they are byte-for-byte identical files under two names. Before relying on the findings operationally, this should be clarified with the course instructor or evidence custodian. Possible explanations include intentional duplication for the laboratory, a labelling/provisioning error, or an intended relationship between the two datasets.

The other recreation/reference files have different hashes and contain sessions from other dates. In particular, one dataset contains a different Gmail account, morcabonilas@gmail.com, with activity dated 25 May 2022. I therefore treated these files as training and methodology-validation material rather than evidence of the same user session.

| Evidence Item                                    | Source Path                                                      | Size | MD5                              | SHA-256 (truncated)                   |
|--------------------------------------------------|------------------------------------------------------------------|------|----------------------------------|---------------------------------------|
| evidence/History_main                            | /media/sf_LabFiles/LabFiles/History                              | 192K | 0c594e9de87348c089554953a67e2c74 | b991b0fa69662c44...cf51bcde3fd19      |
| evidence/Chrome_History_Cryptocurrency_Lab.db    | /media/sf_LabFiles/LabFiles/Chrome_History_Cryptocurrency_Lab.db | 192K | 0c594e9de87348c089554953a67e2c74 | b991b0fa69662c44...cf51bcde3fd19      |
| evidence/Chrome_History_Recreation_Craigslist    | .../Chrome_History_Recreation_Craigslist                         | 160K | a70f8d20d5c7740100a98eb7a2d2ebf6 | 2473b297dddb51f3...eaf871cba8b8e      |
| evidence/Chrome_History_Recreation_Gmail         | .../Chrome_History_Recreation_Gmail                              | 160K | b7a406c9487cd590486e292c5da195f9 | deccal1f2557abd0d..889aa6024237695b4  |
| evidence/Chrome_History_Recreation_Gmail_Headers | .../Chrome_History_Recreation_Gmail_Headers                      | 192K | 7206b9718c710b95ef52408122bbd845 | f63c9d718b8c78fc..3c5d23f49b77d0      |
| evidence/Chrome_History_Recreation_Imgur         | .../Chrome_History_Recreation_Imgur                              | 192K | 32f23e0a7a67fd99ff52d66dac4690f7 | d9b3974ff18c0c53..354fbb9e98ddb88b    |
| evidence/Chrome_History_to_Database_Demo         | .../Chrome_History_to_Database_Demo                              | 160K | 1bf62feb7b4534a2c289418ad149d9f2 | 678ba71cc646f834...322f4fa79f0fdb49b3 |

## 4. Tools, Versions and Methodology

### 4.1 Environment and Tools

- Forensic workstation: Kali Linux rolling release with the zsh shell.

- Database analysis: sqlite3 command-line interface, using the -readonly option for evidence and working-copy queries.
- Data processing: Python 3 with sqlite3, csv, and pandas.
- Hash verification: md5sum and sha256sum from GNU coreutils.
- Screen capture: gnome-screenshot and xfce4-screenshooter.

## 4.2 Methodology

- I created a dedicated case directory with separate evidence/, working/, queries/, logs/, and screenshots/ folders.
- I copied and hashed the original evidence before analysis, following the basic principle of preserving the source and examining a working copy.
- I used sqlite3 in read-only mode to reduce the risk of accidentally changing the database during query development.
- Chrome timestamps, stored as microseconds from the 1601-01-01 WebKit/Chrome epoch, were converted to calendar UTC using `datetime(visit_time/1000000 - 11644473600, 'unixepoch')`.
- I decoded Chrome transition values by identifying the core transition type and checking the qualifier bits, including LINK, TYPED, FORM\_SUBMIT, GENERATED, CHAIN\_START, CHAIN\_END, SERVER\_REDIRECT, and CLIENT\_REDIRECT.
- I exported query results to CSV, hashed the outputs, and used Python/pandas to combine the decoded visit records with the downloads table into a time-ordered master timeline.

During the work, I corrected several normal input mistakes while developing the queries: a table name was entered as visit instead of visits, detetime was entered instead of datetime, and transion was entered instead of transition. I also fixed a permissions problem in the queries/ directory with `chmod`. These were working-environment issues and did not alter the evidence files.

## 5. Detailed Findings and Evidential Interpretation

### 5.1 Craigslist Activity

Between 13:54:18 and 13:58:31 UTC, the history records a Google search for “craigslist,” a redirect sequence to `baltimore.craigslist.org`, and a Craigslist login form submission at 13:54:39. The browser then moved through the “post to classifieds” process, including category selection, listing content, location verification, image attachment, and preview. The final submission occurred at 13:58:22.

At 13:58:31, the live advertisement was viewed. It was titled “cheaper than Rx supplements” and appeared in the “health & beauty by owner” category for Baltimore, with posting ID 7473121658 visible in the management-page URL.

There is a roughly three-minute gap between the content/edit submission at 13:55:15 and the location verification step at 13:58:13. The recorded visit-duration values, including a 177-second dwell time on the edit step, are consistent with the user spending that time preparing the listing.

## 5.2 Gmail Correspondence

After the Craigslist activity, the user searched for and logged into Gmail using the account unsub.fscs@gmail.com. The inbox was opened at 13:59:31. At 14:04:12, the history shows the thread “cheaper than Rx supplements – order” being opened, followed by reply-composition activity between 14:04:35 and 14:04:46.

After about 51 minutes without recorded browsing activity, another message in the same conversation was opened at 14:56:47. Its subject contained the Bitcoin transaction ID:

517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5. This is the same transaction identifier later checked on blockchain explorers.

## 5.3 Imgur Link and File Download

At 14:56:52, immediately after the transaction-related email was opened, the browser followed a Gmail-generated safe-link redirect to Imgur. The redirect URL contains “source=gmail,” which links the navigation to the email context. The browser then opened [imgur.com/dTgrkP7](https://imgur.com/dTgrkP7).

Six seconds later, at 14:56:58, the downloads table records `proof_of_payment.png` being saved to `C:\Users\FSCS_User\Desktop\`. The download record lists [imgur.com](https://imgur.com) as both the referrer and the relevant tab URL. The close timing and matching URLs provide strong support that the downloaded image came from the link opened from the email.

The filename `proof_of_payment.png` suggests that the image was presented as evidence that payment had been made. However, the actual image contents were not available for examination, so I cannot independently confirm what the image showed.

## 5.4 Blockchain and Cryptocurrency Exchange Activity

At 14:58:59, the user searched for “blockchain explorer” and opened Blockchain.com Explorer. At 14:59:06, the transaction ID from the email was searched, and at 14:59:21 the Bitcoin address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt was viewed.

At 14:59:38, the user searched for and signed into Kraken. A device-approval/two-factor authentication page was loaded at 15:00:21. Trade, instant-buy, and funding pages were then visited. At 15:01:21, the Kraken funding page showed a balance of \$41,543.80.

Between 15:01:31 and 15:01:33, the browser followed a link generated by Kraken itself to mempool.space, where the same transaction ID was displayed. At 15:02:05, the Kraken ledger/history page showed \$41,563.70, which is \$19.90 higher than the earlier balance.

The Kraken-generated redirect to the same transaction ID is the strongest link between the transaction discussed in the email and the cryptocurrency exchange activity. It shows that the same transaction was referenced from within the Kraken interface. However, the available history does not contain a transaction-level ledger entry proving that the \$19.90 increase came from that specific Bitcoin transaction. I therefore treat the balance change as an observation, not as proof of settlement.

The user returned to Gmail at 15:02:32 and continued composing a reply in the transaction-related thread.

## 5.5 Comparison / Reference Databases

Chrome\_History\_Recreation\_Craigslist contains a similar Craigslist login-and-post workflow dated 17 June 2022, with posting ID 7497531817. It is about two months later than the primary session and uses a different posting token.

Chrome\_History\_Recreation\_Gmail and Chrome\_History\_Recreation\_Gmail\_Headers were queried separately and are dated differently from the primary artifact. Chrome\_History\_Recreation\_Imgur, despite its filename, contains a Gmail search-and-login sequence dated 25 May 2022 under morcabonilas@gmail.com. This appears to be a labelling discrepancy in the supplied reference set.

Chrome\_History\_to\_Database\_Demo was hashed and catalogued but was not queried as part of this examination, so its contents remain unknown.

For these reasons, I used the reference databases mainly to validate the analysis approach and transition-code decoding. I did not treat them as records of the same user session.



## 6. Integrated Timeline

The following timeline summarizes the master timeline created from the decoded visit records and downloads table. The timestamps are converted from the Chrome artifact and are treated as UTC, subject to the timezone limitations described in Section 7.

| Time<br>(UTC, as<br>converted) | Event Type        | Description                                                                                                                 |
|--------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 13:54:18                       | Search            | Google search for “craigslist” entered in omnibox                                                                           |
| 13:54:19                       | Navigation        | craigslist.org → geo.craigslist.org → baltimore.craigslist.org (redirect chain)                                             |
| 13:54:22                       | Navigation        | accounts.craigslist.org login page loaded                                                                                   |
| 13:54:39                       | Form Submit       | Craigslist account login credentials submitted                                                                              |
| 13:55:03–<br>13:55:05          | Navigation        | Returns to baltimore.craigslist.org, begins “post to classifieds” flow (post.craigslist.org)                                |
| 13:55:10                       | Form Submit       | Posting category selected and submitted                                                                                     |
| 13:55:15                       | Form Submit       | Posting content/description submitted (edit step)                                                                           |
| 13:58:13                       | Form Submit       | Posting location verified (geoverify) ~3 min gap suggests ad text composition                                               |
| 13:58:17                       | Form Submit       | Image attached to posting (editimage)                                                                                       |
| 13:58:19                       | Form Submit       | Posting preview submitted                                                                                                   |
| 13:58:22                       | Form Submit       | Posting finalized/submitted                                                                                                 |
| 13:58:27                       | Navigation        | Posting management page loaded Craigslist post ID 7473121658 assigned                                                       |
| 13:58:31                       | Navigation        | Live advertisement viewed: “cheaper than Rx supplements” (health & beauty, Baltimore)                                       |
| 13:58:38                       | Search            | Google search for “gmail”                                                                                                   |
| 13:58:39–<br>13:59:29          | Navigation        | Gmail navigated to; Google account login flow completed for unsub.fscs@gmail.com                                            |
| 13:59:31                       | Navigation        | Gmail inbox opened “Inbox (12)”                                                                                             |
| 14:04:12                       | Email Viewed      | Message thread opened: “cheaper than Rx supplements order” (buyer inquiry)                                                  |
| 14:04:35–<br>14:04:46          | Email<br>Composed | Reply drafted/composed within the order thread                                                                              |
| 14:04:52–<br>14:05:59          | Navigation        | Returns to thread, then to inbox                                                                                            |
| 14:56:47                       | Email Viewed      | New message opened in same conversation: subject includes Bitcoin TXID 517b2156...88afdc5 (~51 min gap from prior activity) |

|                       |                     |                                                                                                                                               |
|-----------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 14:56:52              | Link Followed       | Link opened from within the Gmail message (URL parameter “source=gmail”) to imgur.com/dTgrkP7, titled “Imgur: the magic of the internet”      |
| 14:56:58              | File Download       | proof_of_payment.png downloaded to Desktop; referrer imgur.com, tab URL imgur.com/dTgrkP7                                                     |
| 14:58:59–<br>14:59:01 | Search / Navigation | Google search “blockchain explorer” → blockchain.com/explorer                                                                                 |
| 14:59:06              | Verification        | Transaction 517b2156...88afdc5 looked up on Blockchain.com Explorer                                                                           |
| 14:59:21              | Verification        | Address 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt viewed on Blockchain.com Explorer                                                                  |
| 14:59:38–<br>14:59:47 | Search / Navigation | Google search “kraken” → kraken.com sign-in page                                                                                              |
| 15:00:21              | Navigation          | Kraken device-approval (2FA) page loaded                                                                                                      |
| 15:01:11–<br>15:01:12 | Navigation          | Kraken trade and instant-buy pages loaded                                                                                                     |
| 15:01:21              | Balance Observed    | Kraken funding page shows balance of \$41,543.80 USD                                                                                          |
| 15:01:31–<br>15:01:33 | Verification        | Link followed from within Kraken’s own interface (kraken.com/redirect) to mempool.space, resolving to the SAME transaction 517b2156...88afdc5 |
| 15:02:05              | Balance Observed    | Kraken ledger/history page shows balance of \$41,563.70 USD (+\$19.90 from prior reading)                                                     |
| 15:02:32–<br>15:02:56 | Email Composed      | Returns to Gmail; further reply drafted within the TXID-referenced thread                                                                     |

## 7. Alternative Explanations, Limitations and Confidence Assessment

### 7.1 Alternative Explanations

- The \$19.90 increase in the Kraken balance may have been an unrelated deposit, a currency-conversion or fee adjustment, or a page-refresh effect. The available evidence does not reconcile the two balance readings to a transaction-level ledger entry.
- Looking up a transaction ID or Bitcoin address on a public blockchain explorer does not prove that the funds were received by the Kraken account. Anyone can search a public transaction.
- The overall sequence could represent a legitimate sale or a payment-verification scam. The available browser history alone cannot distinguish between these explanations.

## 7.2 Limitations

- Only Chrome History data, including visits, URLs, and downloads, was examined. Browser cache, cookies, the downloaded image itself, email server logs, and Kraken account records were not available.
- Chrome timestamps were converted using the standard epoch formula and assumed to be UTC. The source computer's clock was not independently checked against a trusted time source.
- The purpose and evidentiary role of the supplied recreation/lab databases were not fully documented. The identical hashes of History\_main and Chrome\_History\_Cryptocurrency\_Lab.db should be clarified.
- Chrome\_History\_to\_Database\_Demo was not queried, so its contents remain unknown.
- Screenshot files referenced in the examination log were recorded by filename but were not independently reviewed when preparing this report.
- No independent user/device attribution evidence was available. The “FSCS\_User” Windows profile name in the download path is not enough to identify the person who operated the browser.

## 7.3 Confidence Assessment

- HIGH confidence: the existence, order, and timing of the recorded browsing, form-submission, and download events, supported by the verified hash continuity from the source file through the working copy and query outputs.
- MODERATE-TO-HIGH confidence: the connection between the 14:56:47 email, the Imgur link, and the downloaded file, and the repeated appearance of the same transaction ID in the email, Blockchain.com, and the Kraken-generated link.
- MODERATE confidence: describing the session as either a legitimate sale or a fraud pattern, because the additional evidence needed to distinguish the two was not available.
- LOW confidence: attributing the activity to a specific named individual.

## 8. Conclusion

Based on my examination of the History\_main Chrome History artifact, I found a coherent browsing session on 19 April 2022. The session shows the creation of a Craigslist advertisement, Gmail communication with a prospective buyer, receipt and download of an image presented as proof of payment, verification of a Bitcoin transaction on public blockchain explorers, and access to a Kraken account where the displayed balance changed during the same period.

The evidence is internally consistent and the repeated transaction ID provides useful corroboration across the browser activity. In particular, the Kraken-generated redirect to mempool.space is a strong connection between the transaction referenced in the email and the activity inside the exchange interface.

However, the artifacts examined do not establish whether the payment was genuine, partial, or fraudulent. They also do not establish the real-world identity of the person using the browser. I would therefore avoid making a stronger conclusion than the evidence supports.

Before the findings are used for an operational or formal purpose, I recommend the following follow-up steps:

- clarify with the evidence custodian why History\_main and Chrome\_History\_Cryptocurrency\_Lab.db have identical hashes;
- obtain Kraken account statements or transaction-level records to determine whether the \$19.90 balance change relates to the identified transaction;
- examine the actual proof\_of\_payment.png file; and
- obtain the full email headers and, where available, server-side records for the messages identified in Section 5.2.

## 9. Appendices

### Appendix A Examination Activity Log (Directory/File Setup)

Reconstructed from the recorded shell session. Times shown are local workstation time as recorded by the screen-capture utility, and are distinct from the UTC-converted artifact timestamps used in Sections 5–6.

- Case directory created: /home/kali/CIP-B105-CS1-C11-26-DFIT-173065-NAAIRSU-LAWAL, with subfolders evidence/, working/, queries/, logs/, screenshots/ (an earlier attempt using the spelling “NASIRU-LAWAL” failed with “no such file or directory” and was corrected).
- Seven files copied from /media/sf\_LabFiles/LabFiles into evidence/ (History renamed to History\_main on copy).
- MD5 and SHA-256 computed for all evidence files (logs/md5\_evidence.txt, logs/sha256\_evidence.txt) recorded twice during the session with identical results, confirming stability.
- History\_main copied to working/History\_main.sqlite; SHA-256 re-verified as matching.
- File type confirmed via `file`: SQLite 3.x, last written with SQLite 3.36.0, 41 database pages.

- Schema enumerated via ``.tables`` and ``.schema``; 16 tables identified (clusters, clusters\_and\_visits, content\_annotations, context\_annotations, downloads, downloads\_reroute\_info, downloads\_slices, downloads\_url\_chains, keyword\_search\_terms, meta, segment\_usage, segments, sqlite\_sequence, typed\_url\_sync\_metadata, urls, visit\_source, visits).
- Fourteen SQL/Python queries executed against the working copy and comparison databases, exported to queries/ as CSV (01, 05–16, master\_timeline) — see Appendix D for the exact statements.
- A permissions issue on the queries/ directory (permission denied on write) was resolved with ``chmod u+rw`` queries' after an unsuccessful ``sudo -R`` attempt; this is an administrative/environment issue, not an evidence-integrity issue.
- All query outputs hashed (logs/md5\_queries.txt, logs/sha256\_queries.txt).
- Thirteen screen captures taken throughout the session documenting each stage (Appendix E).

## Appendix B Evidence Register

See Section 3 for the summarized register. Full, untruncated hash values are reproduced in Appendix C.

## Appendix C Hash Verification Logs

logs/md5\_evidence.txt / logs/sha256\_evidence.txt (evidence directory, as recorded):

```
0c594e9de87348c089554953a67e2c74 evidence/Chrome_History_Cryptocurrency_Lab.db
a70f8d20d5c7740100a98eb7a2d2ebf6 evidence/Chrome_History_Recreation_Craigslist
b7a406c9487cd590486e292c5da195f9 evidence/Chrome_History_Recreation_Gmail
7206b9718c710b95ef52408122bbd845 evidence/Chrome_History_Recreation_Gmail_Headers
32f23e0a7a67fd99ff52d66dac4690f7 evidence/Chrome_History_Recreation_Imgur
1bf62feb7b4534a2c289418ad149d9f2 evidence/Chrome_History_to_Database_Demo
0c594e9de87348c089554953a67e2c74 evidence/History_main
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
evidence/Chrome_History_Cryptocurrency_Lab.db
2473b297dddb51f3da1ac8298ac6dd482f6d6ee458ce8cb006eacf871cba8b8e
evidence/Chrome_History_Recreation_Craigslist
```

decca1f2557abd0d21627710fbbbbbcc0120749b845f9850889aa6024237695b4

evidence/Chrome\_History\_Recreation\_Gmail

f63c9d718b8c78fcffaea5b1291f9fa145440347c74274f66f3c5d23f49b77d0

evidence/Chrome\_History\_Recreation\_Gmail\_Headers

d9b3974ff18c0c530112689e1e8688cd8c0d0fdf12904eaa354fbb9e98ddb88b

evidence/Chrome\_History\_Recreation\_Imgur

678ba71cc646f83475575be0dbe6864f4efc436a5a021b322f4fa79f0fdb49b3

evidence/Chrome\_History\_to\_Database\_Demo

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History\_main

Working-copy verification:

sha256sum working/History\_main.sqlite

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History\_main.sqlite  
(MATCH)

Query-output hashes (logs/md5\_queries.txt, logs/sha256\_queries.txt) were also captured for all fourteen exported CSV files, recording their state at time of export; values are retained in the case logs and are available on request.

## Appendix D Commands and Queries Reference

Representative commands executed (working directory: case root unless noted; sqlite3 invoked with -readonly throughout):

```
md5sum evidence/* | tee logs/md5_evidence.txt
```

```
sha256sum evidence/* | tee logs/sha256_evidence.txt
```

```
cp evidence/History_main working/History_main.sqlite
```

```
sqlite3 -readonly working/History_main.sqlite ".tables" / ".schema"
```

01\_visits.UTC.csv:

```
SELECT v.id, datetime(v.visit_time/1000000-11644473600,'unixepoch') AS visit_utc,
```

```
    u.url, u.title, v.transition, v.from_visit, v.visit_duration
```

```
FROM visits v JOIN urls u ON v.url = u.id ORDER BY v.visit_time;
```

05\_decoded\_transitions.csv (Python, core-type/qualifier decode of the transition bitmask):

```
CORE
{0:'LINK',1:'TYPED',2:'AUTO_BOOKMARK',3:'AUTO_SUBFRAME',4:'MANUAL_SUBFRAME',
5:'GENERATED',6:'START_PAGE',7:'FORM_SUBMIT',8:'RELOAD',9:'KEYWORD',10:'KEYWORD_G
ENERATED'}

QUALS = [(0x00800000,'BLOCKED'), (0x01000000,'FORWARD_BACK'),
(0x02000000,'FROM_ADDRESS_BAR'),
(0x04000000,'HOME_PAGE'), (0x08000000,'FROM_API'), (0x10000000,'CHAIN_START'),
(0x20000000,'CHAIN_END'), (0x40000000,'CLIENT_REDIRECT'),
(0x80000000,'SERVER_REDIRECT')]

core = CORE.get(t & 0xFF, f'UNK({t & 0xFF})'); quals = '+'.join(n for b,n in QUALS if t & b)
```

07\_craigslist\_activity.csv / 08\_craigslist\_form\_submits.csv:

```
SELECT v.id, datetime(...) AS visit_utc, u.url, u.title,
v.transition & 0xFF AS core_type, v.from_visit
FROM visits v JOIN urls u ON v.url = u.id
WHERE u.url LIKE '%craigslist%' [AND (v.transition & 0xFF) = 7] ORDER BY v.visit_time;
```

10\_gmail\_activity.csv: as above with WHERE u.url LIKE '%mail.google.com%'.

13\_imgur\_activity.csv: as above with WHERE u.url LIKE '%imgur.com%' OR u.url LIKE '%url?q=%imgur%'.

14\_downloads\_full.csv:

```
SELECT id, current_path, referrer, tab_url,
datetime(start_time/1000000-11644473600,'unixepoch') AS start_utc FROM downloads;
```

16\_blockchain\_kraken\_activity.csv: as above with WHERE u.url LIKE '%blockchain%' OR u.url LIKE '%mempool%' OR u.url LIKE '%kraken%'.

master\_timeline.csv: assembled in Python/pandas by concatenating 05\_decoded\_transitions.csv with 14\_downloads\_full.csv (downloads relabelled core\_type = DOWNLOAD, url set from tab\_url), then sorting on the converted UTC timestamp.

09\_recreation\_craigslist.csv, 11\_recreation\_gmail.csv, 12\_recreation\_gmail\_headers.csv, 15\_recreation\_imgur.csv: equivalent queries executed against the corresponding “recreation” database files in place of History\_main.sqlite.

Documented input corrections during development: table name “visit” corrected to “visits”; function name “detetime” corrected to “datetime”; column name “trasition” corrected to “transition”. A queries/ directory permission error (“permission denied” on redirected output) was resolved with `chmod u+rwX queries`.

## Appendix E Screenshot Register

The following screen captures were recorded during the examination (filenames and local workstation capture times as logged; not independently reviewed for this report).

| Local Time | Filename                          | Context (inferred from surrounding log)             |
|------------|-----------------------------------|-----------------------------------------------------|
| 18:12:45   | 01_terminal.png (initial attempt) | gnome-screenshot install/first capture              |
| 18:20:51   | 001_terminal.png                  | Evidence hashing (md5/sha256) captured              |
| 18:32:51   | 002_terminal.png                  | History_main.sqlite copy and schema dump            |
| 18:49:34   | 003_terminal.png                  | Full schema (.schema) review                        |
| 19:26:39   | 004_terminal.png                  | Initial visits-query troubleshooting                |
| 19:27:16   | 005_terminal.png                  | Continued query troubleshooting                     |
| 19:29:35   | Hashes_values.png                 | Hash values captured via xfce4-screenshooter        |
| 19:49:24   | 006_terminal.png                  | Decoded transitions query (05) success              |
| 19:56:56   | 007_terminal.png                  | Craigslist activity query (07) results              |
| 20:04:02   | 008_terminal.png                  | Gmail activity query (10) / Imgur grep              |
| 20:13:58   | 009_terminal.png                  | Downloads and recreation-Imgur queries              |
| 20:15:36   | 010_terminal.png                  | Blockchain/Kraken query (16) and grep of tx/address |
| 20:16:51   | 011_terminal.png                  | Master timeline assembly (pandas)                   |
| 20:22:49   | 012_terminal.png                  | Final hashing of queries/ output                    |

*End of report.*